



Assessment of precipitable water vapor retrieved from precise point positioning with PPP-B2b service

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Abstract

Real-time precise point positioning (RT-PPP) is an effective method to retrieve precipitable water vapor (PWV), which has better timeliness compared with the European Center for Medium-Range Weather Forecasts Reanalysis (ERA5). The real-time service (RTS) required for RT-PPP broadcasted by network link may be affected by outages. The PPP-B2b service of the BeiDou satellite navigation system (BDS) provides RTS via satellite, which is suitable for PWV retrieving without the influence of network issues. However, previous studies have not addressed the assessment of PWV retrieved from PPP-B2b over the continent and ocean environments. Here, we show the retrieval performance of the tropospheric zenith delay (ZTD) and PWV with different RTS products. First, the ZTD accuracy of ERA5 is 16.5 mm, compared with 10.9 mm for the Centre National d'Etudes Spatiales (CNES) solution for the global positioning system (GPS). Meanwhile, the PPP-B2b-GPS is 26.4 mm, while GPS/BDS-3 shows an improvement of 34% over GPS only. Second, the PWV difference between PPP-B2b-GPS and ERA5 is 4.9 mm, and GPS/BDS-3 is 3.4 mm. The agreement between CNES and ERA5 is slightly better than that between PPP-B2b and ERA5, where the PWV accuracy of GPS and GPS-BDS-3 is 2.7 mm and 2.3 mm, respectively. Finally, the PWV using PPP-B2b GPS-BDS-3 for the shipboard experiments is consistent with the CNES results. The difference between them is within 1.5 mm after convergence. We anticipate our assay to be a starting point for applying PPP-B2b service in meteorological studies on land and maritime areas susceptible to network unavailability.

Keywords Assessment · B2b service · Precise point positioning · Precipitable water vapor · Tropospheric zenith delay

Introduction

The global navigation satellite system (GNSS) has the characteristics of global coverage and all-day service, which provides high-precision positioning, navigation and timing,

and essential meteorological information necessary for numerical weather prediction (Zumberge et al. 1997; Gutman et al. 2004; Rohm et al. 2014) and much more (Ferreira et al. 2021). The early works of Askne and Nordius (1987) and Bevis et al. (1972, 1994) used the observations of the global positioning system (GPS) to retrieve the water vapor information and studied the correlation between the

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atmospheric wet delay and atmospheric precipitable water. Furthermore, their studies provided the fundamental principles for calculating precipitable water vapor (PWV) from GPS observations. Rocken et al. (1993) investigated PWV retrieval using GPS and achieved millimeter-level results regarding those based on water vapor radiometer measurements. Precise point positioning (PPP) is an essential technique in GNSS to achieve high-precision positioning and PWV retrieval without reference stations (Kouba and Héroux 2001; El-Mowafy et al. 2017). In a nutshell, the estimation of the PWV relies on the following steps: (1) the correction of the modelable errors, such as satellite orbit errors, clock errors, and antenna phase center offsets, in the PPP equation (Wang et al. 2019); (2) real-time estimation of the zenith total delay (ZTD) by linearizing the positioning equation; and (3) combination of the Saastamoinen model and the conversion factor (Saastamoinen 1972; Shoji et al. 2009; Sun et al. 2021). Alternatively, ZTD and PWV for a given station can be obtained by interpolating the meteorological data of the grid points based on the fifth-generation European Center for Medium-Range Weather Forecasts Reanalysis (ERA5) fields (Jiang et al. 2016; Zhang et al. 2019; Ssenyunzi et al. 2020). ERA5 is the latest atmospheric reanalysis product from the European Center for Medium-Range Weather Forecasts (ECMWF). However, there is a latency of several days for the reanalysis data to be available, making it challenging to meet the demand for real-time weather research.

The emergence of real-time service (RTS) has improved the accuracy of real-time PPP (Hadas and Bosy 2015; Abdelazeem et al. 2016; Elsobeiey and Al-Harbi 2016), and this unique timeliness has opened a new opportunity for PWV retrieval. For instance, previous studies have demonstrated that the PWV accuracy from real-time PPP is within 3 mm, which can be applied to real-time meteorological research (Yuan et al. 2014; Shi et al. 2015; Ahmed et al. 2016). With the development of the BeiDou global navigation satellite system (BDS), Galileo system, and GLONASS, the PPP gradually developed from GPS only to a multi-system combination (e.g., An et al. 2020). The multi-system PPP has a more reasonable spatial distribution of available satellites and a shorter convergence time, especially with improved retrieval accuracy (Li et al. 2015; Lu et al. 2015, 2017). However, a stable retrieval environment is also a concern, and the integrity of the retrieval results is as important as the accuracy. Previously, RTS was broadcasted to the user through the network link by organizations such as the Centre National d'Etudes Spatiales (CNES). Nevertheless, the external environment can affect the network, and the RTS data transmission may be interrupted. Furthermore, the user cannot receive RTS data in remote areas without network coverage. Since the PPP-B2b service emerged based on the geostationary orbit (GEO) satellites, which support BDS-3

and GPS (Yang et al. 2021; Zhang et al. 2021), a new way to overcome the issues with the RTS-based network links has emerged. Due to its solid anti-interference capability, the PPP-B2b service will become a popular standard for PWV retrieval in China and surrounding regions. However, there is a lack of studies comparing the performance of real-time retrieval ZTD and PWV based on PPP-B2b and CNES. To date, the PPP-B2b service is limited to China and its surrounding areas, so our assessment only uses GNSS data from China.

In this study, we assess the performance of real-time retrieval ZTD and PWV based on PPP-B2b through International GNSS Service (IGS) static and shipborne dynamic datasets relative to ERA5 in China. The remainder of this study is structured as follows. Second section describes the method for recovering precision orbits and clock differences for the PPP-B2b service. In addition, the processes of calculating ZTD and PWV using PPP/ERA5 are introduced. The detailed strategy and experimental data of PPP are explained in third section. Fourth section assesses the performance of ZTD and PWV based on static and dynamic data. Fifth section introduces the discussion of experiments. The conclusion of the study is presented in sixth section.

Methodology

The description of the experimental method can be divided into three parts. Firstly, the method of recovering precision satellite orbits and clocks from RTS data is introduced. Secondly, the detailed process of real-time retrieval of ZTD and PWV based on PPP is described. Finally, the method of obtaining PWV from ERA5 data by integration is presented.

Recovery of precise orbits and clock offsets based on PPP-B2b service

The accuracy of satellite orbits and clocks usually affects the convergence time and performance of real-time PPP positioning (Tao et al. 2021). Thus, the relevant corrections provided by the PPP-B2b service become essential. As defined in the Interface Control Document (ICD) of the PPP-B2b service, the orbit message includes corrections of radial, along, and cross directions, and users can combine LNAV and CNAV1 navigation messages to recover precision orbits and clocks (CSNO 2020). While the radial error of orbit is usually at the centimeter level, and the satellite clock error is within 0.2 ns (Nie et al. 2021; Xu et al. 2021). The precise orbits and clocks can be derived as follows:

$$\delta X_b = [e_r \ e_a \ e_c] \delta O \quad (1)$$

$$\begin{bmatrix} e_r \\ e_c \\ e_a \end{bmatrix} = \begin{bmatrix} \frac{r_p}{|r_p|} \\ \frac{r_p \times r_v}{|r_p \times r_v|} \\ e_c \times e_r \end{bmatrix} \quad (2)$$

where δX_b is the orbit correction vector in Earth-center Earth-fixed (ECEF) coordinate system, δO denotes the orbit correction message from the PPP-B2b service, and r_p and r_v are the satellite unit and velocity vectors, calculated by broadcast ephemeris. To recover the real-time precise orbits with δX_b , the following equation can be used:

$$X_{pre} = X_b - \delta X_b \quad (3)$$

In Eq. (3), X_{pre} represents the real-time precise satellite orbit vector, and X_b is the satellite position vector of the broadcast ephemeris. Applying the clock correction to broadcast clock offsets, the real-time precise clock offset vector T_{pre} can be computed as:

$$T_{pre} = T_b - \delta T_b / c \quad (4)$$

where T_b is the clock offset vector derived from the broadcast ephemeris, δT_b is the clock correction of the PPP-B2b service, and the constant c is the speed of light.

Determination of PWV from real-time PPP model

The GNSS code and phase ionospheric-free (IF) combinations can be presented as (Geng et al. 2010; Zhang et al. 2020):

$$\begin{cases} P_{IF} = \rho + c \cdot dt_r - c \cdot dt_s + M \cdot T + \varepsilon_{P_{IF}} \\ L_{IF} = \rho + c \cdot dt_r - c \cdot dt_s + M \cdot T + \lambda_{IF} \cdot N_{IF} + \varepsilon_{L_{IF}} \end{cases} \quad (5)$$

where P_{IF} and L_{IF} are the code and phase measurements with the IF combination, respectively, ρ represents the geometric distance from the GNSS satellite to the receiver, dt_r and dt_s denote the clock offsets of the receiver and satellite, respectively, T is the zenith tropospheric delay, M indicates the related mapping function, N_{IF} is the IF combination phase ambiguity, λ_{IF} is the IF wavelength. $\varepsilon_{P_{IF}}$ and $\varepsilon_{L_{IF}}$ are the code and phase measurement noises, respectively. With the precise orbit and clock offsets, the above equations can be linearized as:

$$\begin{cases} P_{IF} = -e_{rs} \cdot x + c \cdot dt_r + M_w \cdot ZWD + \varepsilon_{P_{IF}} \\ L_{IF} = -e_{rs} \cdot x + c \cdot dt_r + M_w \cdot ZWD + \lambda_{IF} \cdot N_{IF} + \varepsilon_{L_{IF}} \end{cases} \quad (6)$$

where e_{rs} is the unit vector from satellite to receiver, and the x is the position increment of the receiver, and ZWD represents the zenith wet delay.

Here, we use the extended Kalman filter (EKF) to estimate the total zenith tropospheric delay. The zenith hydrostatic delay (ZHD) is obtained by the Saastamoinen model (Wilgan et al. 2015):

$$ZHD = \frac{0.002277 \cdot P_s}{f(\phi, h)} \quad (7)$$

$$f(\phi, h) = 1 - 0.0026 \cdot \cos(2\phi) - 0.00028 \cdot h \quad (8)$$

where P_s is the atmospheric pressure on the station, which is derived from the Global Pressure and Temperature 3 (GPT3) model, $f(\phi, h)$ is the variations of gravity value above the ellipsoid, and ϕ and h are the latitude and height of the station, respectively. The real-time PWV can be calculated through a conversion factor as (e.g., Lu et al. 2015):

$$PWV = Q \cdot (ZTD - ZHD) \quad (9)$$

with

$$Q = \frac{10^6}{\rho_w R_v \left(\frac{k_3}{T_m} + k'_2 \right)} \quad (10)$$

In Eq. (9) Q represents the conversion factor, and ρ_w represents the density constant of liquid water, R_v is the gas constant of water vapor, T_m is the weighted mean temperature, and the constants k_3 and k'_2 are the refractivity constants.

Determination of ZTD and PWV from ERA5 Reanalysis data

ERA5 hourly pressure level dataset contains the required meteorological parameters, such as temperature, pressure, and specific humidity, for estimating ZTD and PWV. Using the meteorological parameters given on grid bins at $0.25^\circ \times 0.25^\circ$, the ZTD and PWV can be computed at each grid bin and then interpolated at a given GNSS site. (Alternatively, the meteorological parameters could be interpolated to each GNSS site then the ZTD and PWV could be computed directly.) Noteworthy, it is necessary to perform height unification and horizontal unification to eliminate the differences between grid point coordinates and station coordinates. Here, the integral principle is used to process the pressure level data to obtain ZTD and PWV. The total refractivity N can be computed based on the following formula (e.g Chen et al. 2011; Wilgan et al. 2015):

$$N = k_1 \frac{p - e}{T} + \left(k_2 \frac{e}{T} + k_3 \frac{e}{T^2} \right) = N_{dry} + N_{wet} \quad (11)$$

$$e = \frac{qp}{0.622 + 0.378q} \quad (12)$$

where $k_1 = 77.604$ K/Pa, $k_2 = 64.79$ K/Pa, and $k_3 = 377,600.0$ K²/Pa, N_{dry} and N_{wet} are the refractivity of dry gas and wet gas, respectively, p denotes the atmosphere pressure, T is the temperature, e is the water vapor pressure, and q is the specific humidity. The ZTD can then be computed as:

$$ZTD = 10^{-6} \sum_{i=1}^n N_{dry_i} \Delta H_i + 10^{-6} \sum_{i=1}^n N_{wet_i} \Delta H_i = ZHD + ZWD \quad (13)$$

where ΔH_i represents the height difference between adjacent meteorological levels. N_{dry_i} and ΔN_{wet_i} are the average refractivity of dry gas and wet gas between adjacent levels, respectively. The PWV can be computed using Eq. (9), or through the following integral:

$$PWV = \frac{1}{\rho_w} \int \frac{q}{g} dp \quad (14)$$

where g denotes the gravity value. The height of the meteorological data level often does not coincide with that of the station. Therefore, the PWV calculated by Eq. (14) needs to be fitted to the station height using an exponential model as (cf. Shi et al. 2022):

$$PWV_s = PWV \cdot \exp\left(-\frac{H_m - H_s}{2000}\right) \quad (15)$$

where H_m is the height of the grid point closest to the station in the vertical direction, and H_s is the station height. The station PWV is obtained by bilinear interpolating the PWV of the grid points around the station to eliminate the horizontal bias.

Experimental data and setup

We tested the performance of the real-time PWV retrieval by processing both static and dynamic data. Considering that the current service range of PPP-B2b is limited to China and surrounding areas (CSNO 2020; Tao et al. 2021; Xu et al. 2021), nine IGS stations located in China were selected (Fig. 1). We chose the IGS station over China for two reasons. First, the current service range of PPP-B2b is limited to China and its surrounding area. Second, the IGS data are free and open to users, which also facilitates the replication of experiments by other researchers. The data

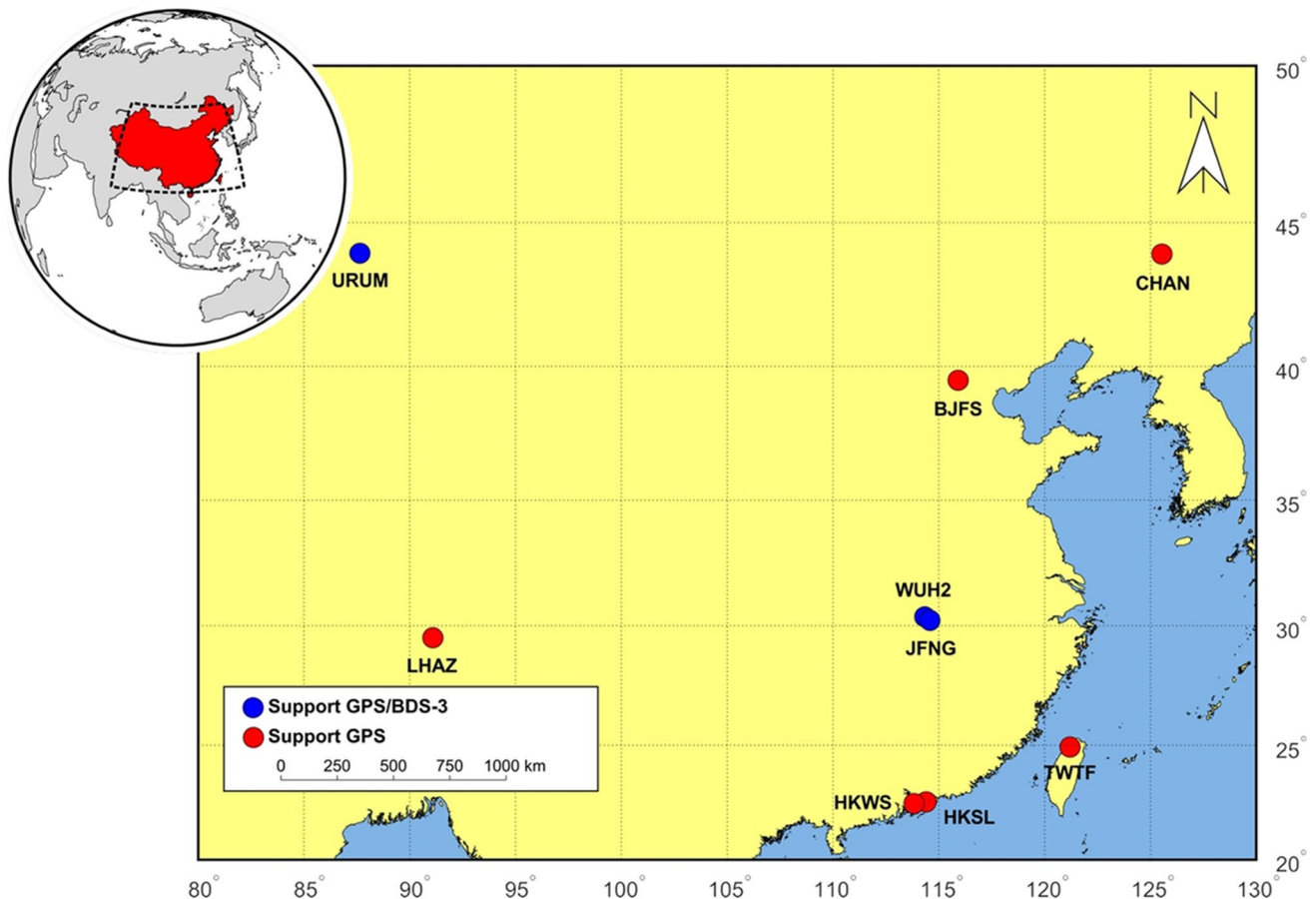


Fig. 1 The distribution of nine IGS stations located in China. The blue circles indicate the stations where the receivers support GPS and BDS-3, and the red circles depict the stations where the receivers only support GPS. The scale refers to the map's center

collection period is from the day of the year (DOY) 195 to DOY 200 in 2021. Furthermore, we used a special receiver to collect PPP-B2b data for the same six days. The IGS data for each day is backed up on the official website and can be downloaded by users for free. However, PPP-B2b data currently does not have corresponding backups on the website. Ordinary users can only receive PPP-B2b data in real-time through specific receivers with support for receiving PPP-B2b data, and most receivers do not yet have this function. (They will be upgraded gradually in the future.) The lack of a large amount of PPP-B2b data availability led us to specific difficulties in DOY chosen (195–200).

Up to now, the PPP-B2b service has supported GPS and BDS-3 satellite navigation systems. However, the present study tests the retrieval performance of GPS only and the GPS-BDS-3 combination. The ERA5 and CNES RTS products are used only for comparison purposes. The horizontal

resolution of ERA5 pressure level data for the corresponding period is $0.25^\circ \times 0.25^\circ$, and the vertical resolution is 37 levels. Table 1 shows the detailed PPP processing strategy.

The processing flow of ZTD/PWV is summarized in Fig. 2. For the ZTD, the ZPD file of IGS provides the precise ZTD for each IGS station, which is used as the actual value to evaluate the retrieval performance of real-time PPP and ERA5.

Although it is recognized that the radiosonde datasets have the highest accuracy, their temporal resolution and location distribution cannot meet the demand of the experimental stations for PWV retrieval. Then the PWV with PPP and ERA5 data is chosen for comparison, and the difference between them is used to assess the accuracy.

Results

This section gives the accuracy of PWV retrieval for static data and the performance in the shipborne dynamic environment.

Comparison of the ZTD of PPP-B2b/CNES retrieved in static mode with ERA5

The receivers do not fully support the reception of BDS-3 data, so the PPP experiments of some stations only use GPS. Figure 3 shows the ZTD distribution of GPS only and ERA5, as the true value of ZTD provided by IGS. It can be seen that the ZTD calculated for different RTS are similar to the IGS in terms of variation trends for different stations (Fig. 3). In addition, the distributions of CNES and ERA5 are closer to the IGS. There are small fluctuations in the ZTD based on PPP-B2b, which are affected by the accuracy of satellite clocks. Note that the experimental

Table 1 Processing strategy

Item	Processing Strategies
Observation type	GPS L1/L2, BDS-3 B1/B3
Combination	Ionosphere-free
Elevation mask	10°
Sampling interval:	30 s
Weight	Elevation-dependent weight
Troposphere	Estimated the ZTD as a random walk noise, while the ZHD is based on the Saastamoinen model
Mapping function	NMF function
Station displacement	Solid earth tide, ocean tide, pole tide
Ambiguity	Float
Estimator	Extended Kalman Filter
Satellite orbit and clock	PPP-B2b/CNES

Fig. 2 The processing flow of ZTD/PWV and accuracy assessment

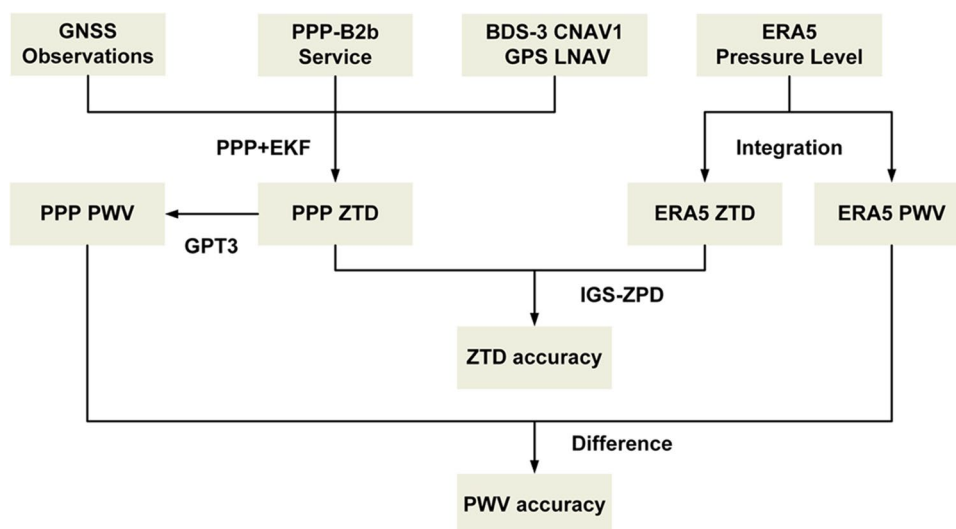
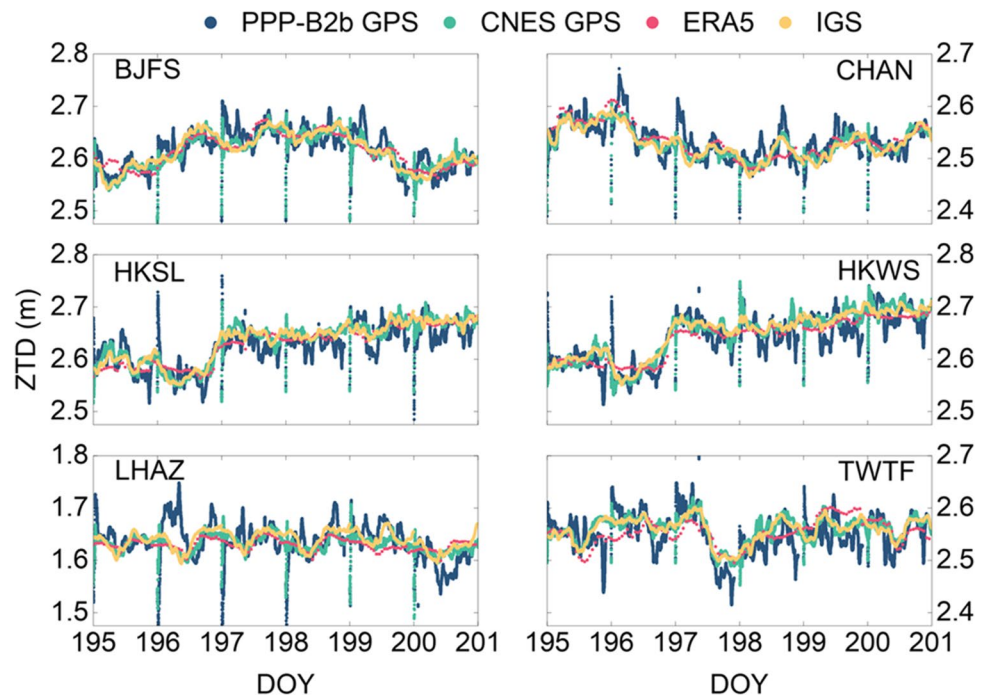


Fig. 3 The dots show the comparison of ZTD based on GPS and ERA5



data were not processed continuously because the RTS of PPP-B2b GPS is available until 10:00 p.m. for each day. This leads to a short convergence process of the ZTD calculated by PPP, which is shown as multiple vertical points with a wide range of fluctuations at the beginning of each day in Fig. 3.

The three stations, JFNG, URUM, and WUH2, are equipped with receivers able to receive both GPS and BDS-3 signals (see Fig. 1 for the respective locations). Figure 4 shows the ZTD of GPS and GPS/BDS-3. It can be seen that the GPS/BDS-3 combination has less fluctuation range than GPS only. This is more obvious in the PPP-B2b results due

Fig. 4 Comparison of ZTD series based on GPS/BDS-3 and ERA5 over three selected IGS stations

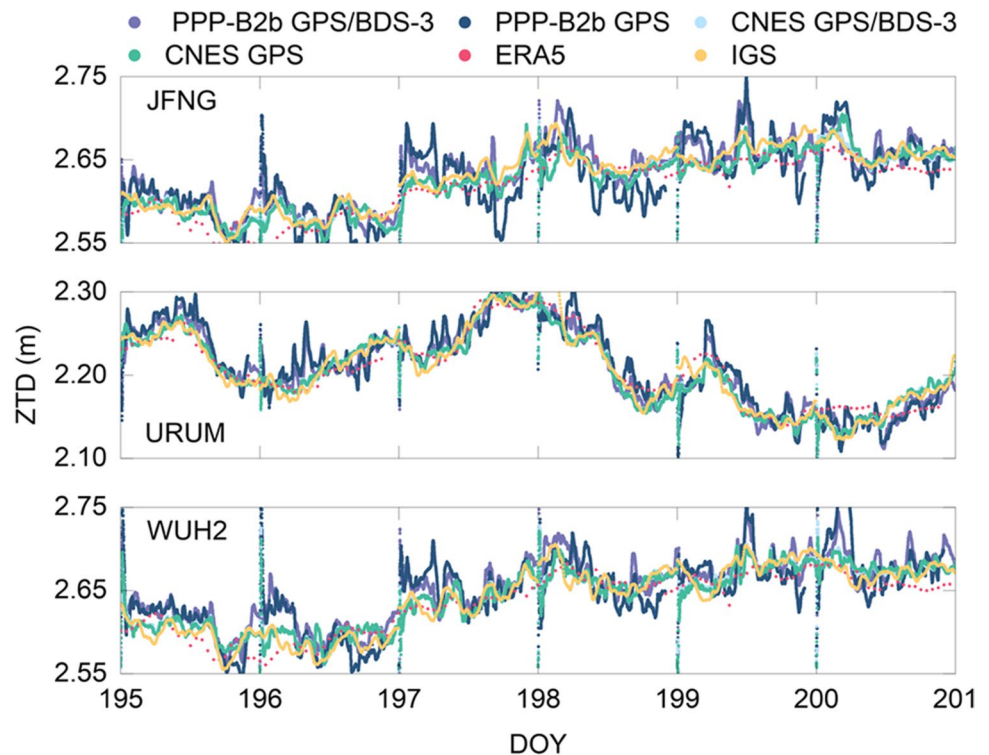
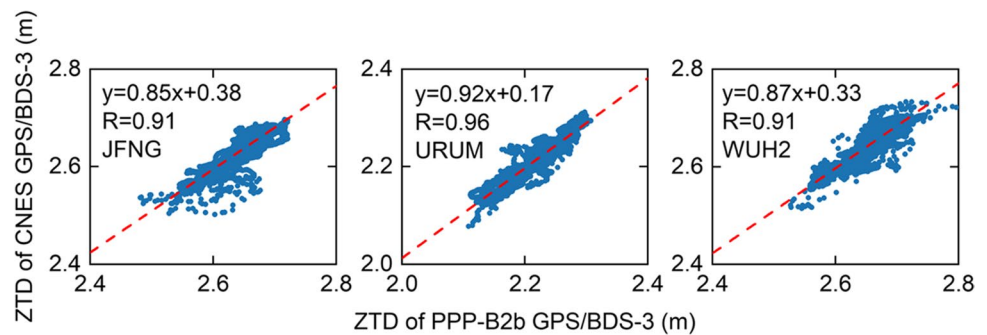


Fig. 5 Correlation analysis between the ZTD of PPP-B2b and CNES



to the better clock accuracy of BDS-3 than GPS in PPP-B2b (Tao et al. 2021).

Figure 5 compares ZTD calculated by the GPS-BDS-3 combination of PPP-B2b at JFNG, URUM, and WUH2 sites regarding their scatter plots and relationship (correlations). Overall, the ZTD calculated by the GPS-BDS-3 combination of PPP-B2b correlates with CNES for the three stations (Fig. 5). The correlation coefficients (R) of all three stations exceed 0.90, indicating a strong correlation between PPP-B2b and CNES.

In order to avoid the influence of the convergence period, which affects the accuracy assessment, the ZTD assessment was considered from 0:30:00 each day. That is, when real-time PPP uses EKF filtering to estimate the ZTD, it takes a while to ensure that the estimated ZTD converges to a given tolerance. This means that the ZTD within the convergence time presents a significant deviation from the actual value. Hence, it is better to exclude the ZTD estimations within the convergence time for assessment purposes; otherwise, the convergence time of real-time PPP would affect the assessment. However, the convergence time does not affect ERA5 because it is based on integration to calculate ZTD ("Determination of ZTD and PWV from ERA5 Reanalysis data" section). So, the effect of convergence time on accuracy evaluation is limited to real-time PPP and not ERA5. Figure 6 shows each IGS station's root-mean-square error (RMSE) values. It is clear that CNES has the best accuracy, followed by ERA5, as summarized by the respective RMSE values. Except for the TWTF, the RMSE of the remaining stations are within 20 mm for CENS and EAR5. For the PPP-B2b service, the accuracy of GPS/BDS-3 is close to that of CNES, but the maximum error is over 30 mm for GPS only.

Table 2 gives the mean RMSE for the nine IGS stations. The overall accuracy of CNES is slightly higher than that of PPP-B2b, while the RMSE of ERA5 lies between PPP-B2b and CNES at 16.5 mm. It is worth mentioning that in PPP-B2b, the GPS/BDS-3 combination achieves a 9 mm accuracy improvement to GPS only.

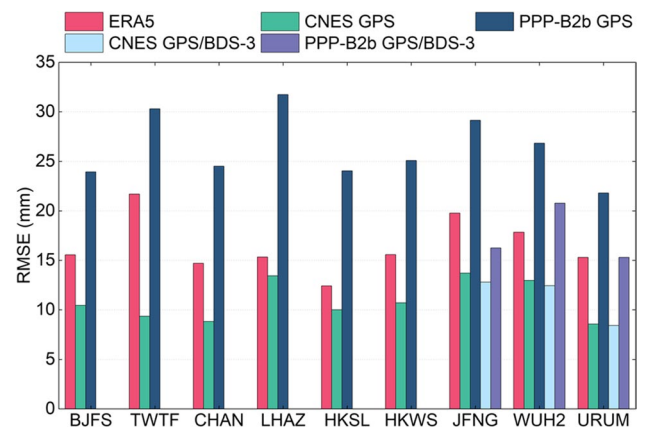


Fig. 6 ZTD accuracy for each station regarding RMSE

Table 2 Mean RMSE of all stations

ZTD accuracy	ERA5	CNES GPS	PPP-B2b GPS	CNES GPS/BDS-3	PPP-B2b GPS/BDS-3
Mean RMSE (mm)	16.5	10.9	26.4	11.2	17.4

Comparison of the PWV retrieved from a static mode of the PPP-B2b/CNES with ERA5

The weighted mean temperature, T_m , is an important parameter for PWV retrieval in real-time PPP. We mainly used the GPT3 model to obtain the real-time T_m , which is less accurate than using meteorological sensors. Figure 7 illustrates the T_m differences between GPT3 and ERA5 at which the T_m values retrieved from ERA5 were used as references. Most of them are within 8 K, indicating that the GPT3 model can be applied to real-time retrieval.

Since the GPT3 model requires the real-time coordinates of the stations, Fig. 8 shows the effect of different RTS to calculate the real-time coordinates on T_m . It can be seen that the differences among all four solutions (i.e.,

Fig. 7 Differences of T_m between GPT3 and ERA5

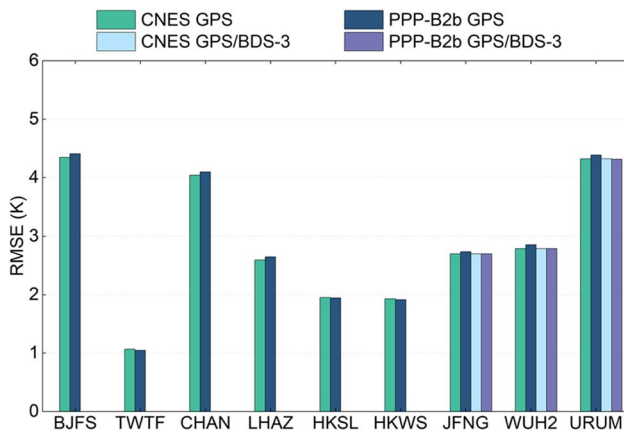
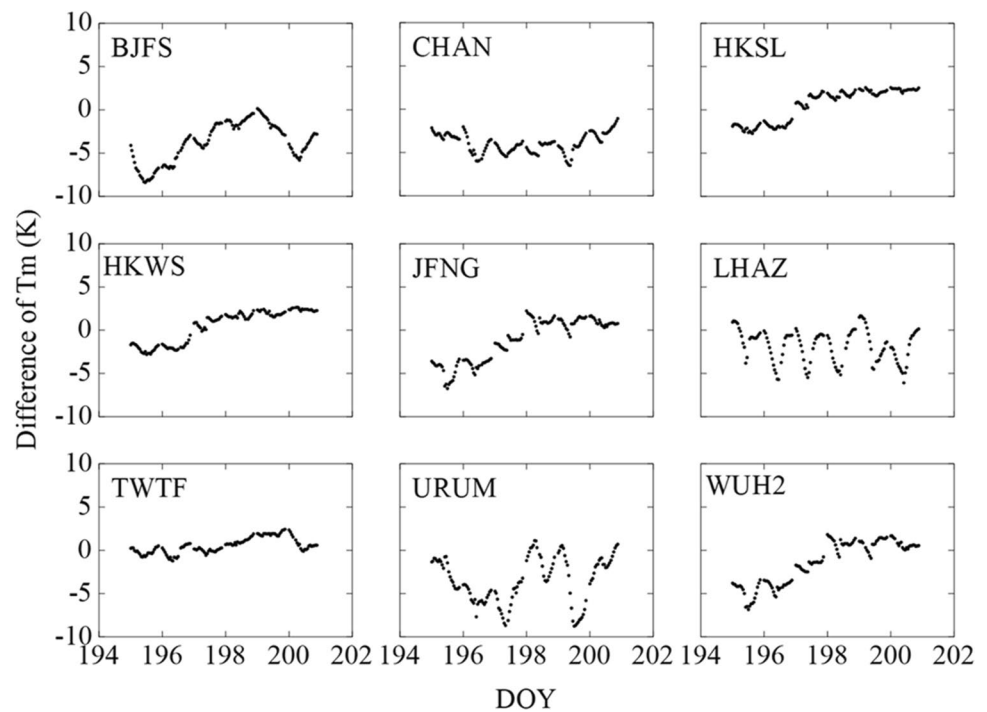


Fig. 8 The RMSE values of T_m

CNES GPS, PPP-B2b GPS, CNES GPS/BDS-3, PPP-B2b GPS/BDS-3) are almost negligible.

Figure 9 shows the results of PWV retrieval. Taking the PWV of ERA5 as the reference, the bias range of PPP-B2b is slightly higher than that of CNES and exceeds 5 mm in a few epochs. The trend of PWV calculated by PPP-B2b and CNES can be consistent with that of ERA5.

Daily total rainfall data observed at the Hong Kong Observatory meteorological station, which is located within 30 km of the HKSL and HKWS stations, were used for inter-comparison purposes. Figure 10 shows that the total rainfall

exceeds 70 mm at DOY 200. The rise of PWV at the Hong Kong Observatory meteorological station may be associated with heavy rainfall. Ignoring the distance bias, the PWV of HKSL and HKWS from PPP-B2b also reflects this rising trend. This shows the potential of the PPP-B2b service for PWV retrieval, rainfall forecasting, and other meteorological monitoring.

Figure 11 shows the retrieval PWV of GPS/BDS-3 by IGS stations. CNES has slightly better accuracy and convergence time than PPP-B2b. For the PPP-B2b, the results of GPS/BDS-3 show some improvement over GPS only, and this improvement is not apparent in CNES.

Figure 12 shows the distribution of PWV difference. It can be seen that for the CNES, most differences are concentrated within ± 4 mm of either GPS or GPS/BDS-3. In contrast, the distribution of GPS for PPP-B2b is concentrated within ± 6 mm, and that of GPS/BDS-3 is more within ± 4 mm.

Figure 13 presents the comparison between PWV based on PPP-B2b and CNES at JFNG, URUM, and WUH2 stations regarding their scatter plots and correlations. As can be seen, there is still a satisfactory correlation between the PWV calculated by the GPS/BDS-3 combination of PPP-B2b and that of CNES, with all correlation coefficients above 0.9 (Fig. 13).

The RMSE values of PWV are given in Fig. 14. For the RMSE of GPS, the maximum difference between CNES and PPP-B2b does not exceed 4 mm, while the difference narrows to within 2 mm in the GPS/BDS-3 combination.

Fig. 9 Comparison of ZTD retrieved from GPS and ERA5

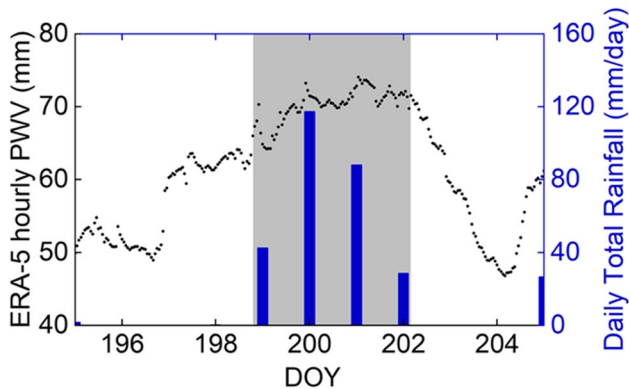
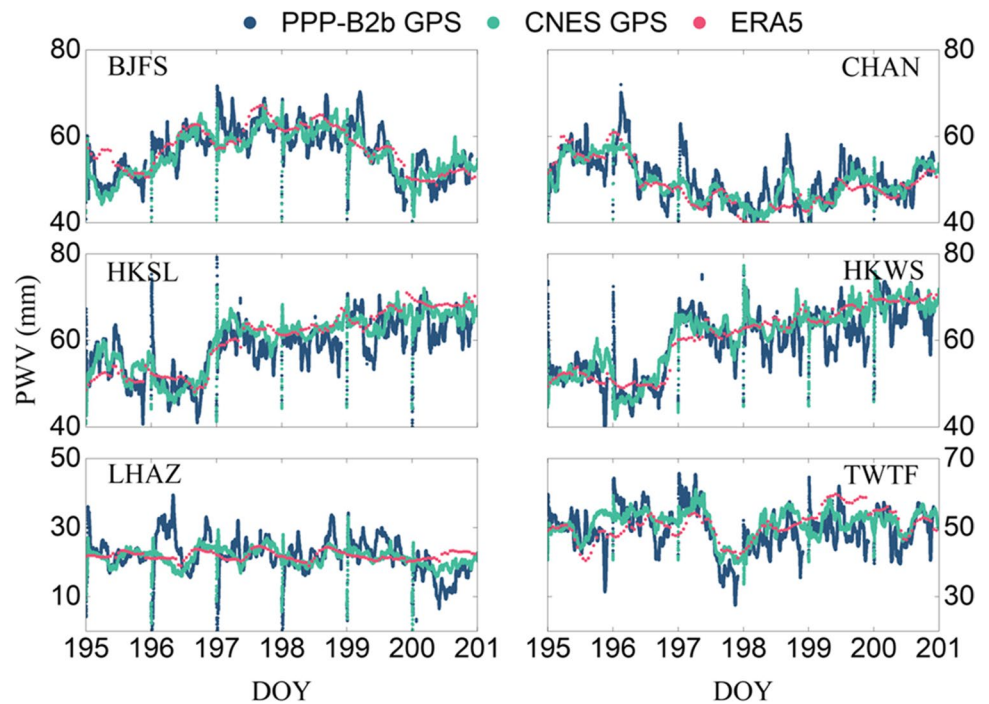


Fig. 10 Hourly PWV and daily rainfall at the Hong Kong Observatory meteorological station

Table 3 shows the mean accuracy of all stations. The RMSE of the PPP-B2b GPS/BDS-3 combination improves by about 1.5 mm over GPS only. Meanwhile, the RMSE of the CNES GPS/BDS-3 combination also improves by about 0.4 mm over GPS only.

Retrieval experiments of dynamic shipboard data

The first set of shipboard GPS data was collected from 4:30:00 to 8:30:00 on August 4, 2021, and was located in Houcha Bay, Qingdao. The shipboard receiver is shown in Fig. 15a. In addition to the station environment changing

from static to dynamic, the signal reflected from the seawater surface increases the multipath effect. The multipath errors of GPS data are shown in Fig. 15b, and the multipath RMS of the L1 signal is 0.376 m, while L2 reaches 0.379 m.

Due to the lack of meteorological sensors to provide true PWV values over the ship, the PWV of the coordinates estimated by real-time PPP was calculated by ERA5 as a reference. The positioning accuracy of real-time PPP is usually in the centimeter to decimeter range, and the variation of PWV at this horizontal distance difference can be neglected. Figure 16 shows the PWV retrieval of PPP-B2b and CNES, and the red points indicate ERA5. The PWV based on the CNES solution is closer to ERA5 than PPP-B2b and has a faster convergence time. This should be related to the difference in the accuracy of their clocks.

The second set of shipboard GPS/BDS-3 data was collected from 3:00:00 to 5:55:00 on December 21, 2021, and was located in Tangdao Bay, Qingdao. The shipboard receiver with a choke ring antenna is shown in Fig. 17a. Figure 17b shows that the multipath of the GPS has improved significantly due to the choke ring antenna, reducing the RMS to 0.125 m for L1 and 0.222 m for L2.

Figure 18 shows the results of GPS and GPS/BDS-3, with the red points indicating the PWV of ERA5. For PPP-B2b, the convergence speed of GPS/BDS-3 is significantly better than that of GPS only, and the difference from the PWV of CNES is mostly within 1.5 mm. This indicates that GPS/BDS-3 can achieve similar retrieval performance as the CNES in a dynamic environment.

Fig. 11 Comparison of PWV based on GPS/BDS-3 and ERA5

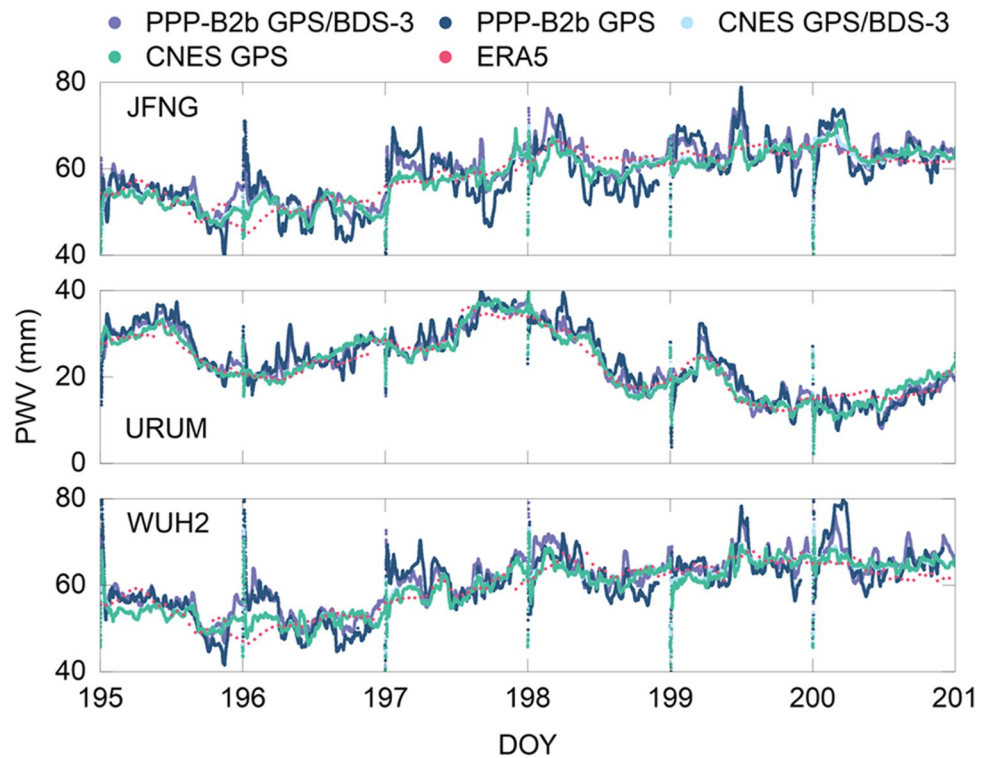


Fig. 12 Distribution of PWV differences

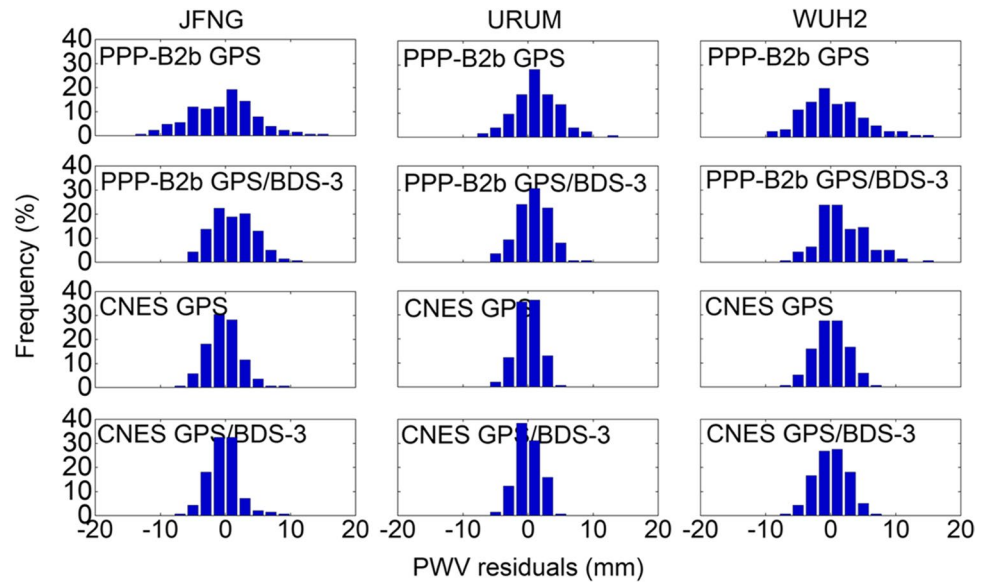
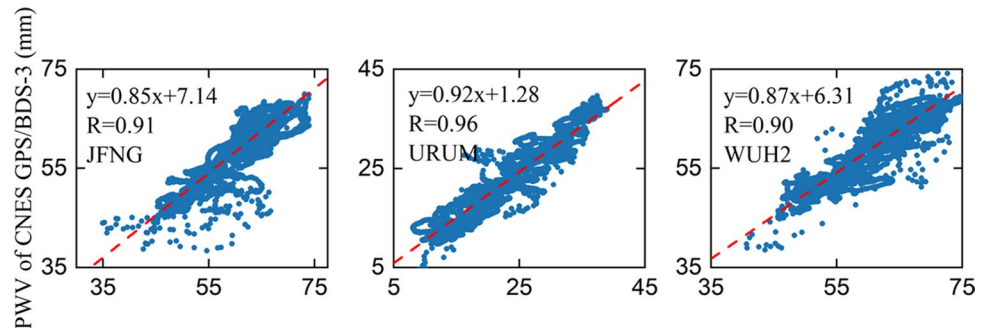


Fig. 13 Correlation analysis between the PWV of PPP-B2b and CNES



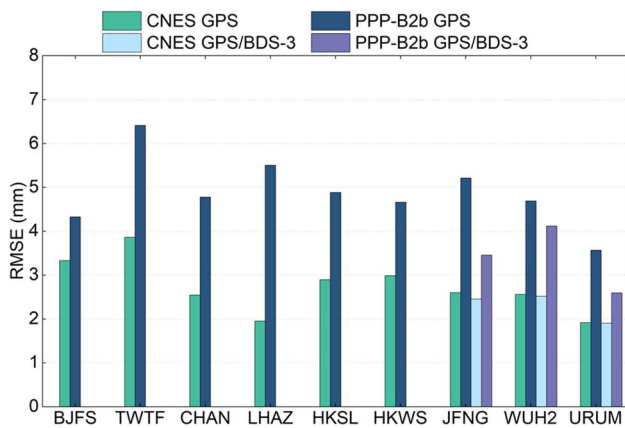


Fig. 14 PWV accuracy of each station

Table 3 Average RMSE of all stations

PWV accuracy	CNES GPS	PPP-B2b GPS	CNES GPS/BDS-3	PPP-B2b GPS/BDS-3
Mean RMSE (mm)	2.7	4.9	2.3	3.4

Discussion

The main goal of this study was to assess the performance of real-time retrieval ZTD and PWV based on PPP-B2b through IGS static and shipborne dynamic datasets relative to ERA5. Real-time PPP is an effective method to retrieve PWV, which has better timeliness than the ERA5. The PPP-B2b service of the BDS provides RTS corrections based on GEO satellites, which offer strong anti-interference ability. Previous studies have not evaluated PWV retrieved from PPP-B2b over different environments. Overall, our results show that the PWV difference between PPP-B2b GPS and ERA5 is 4.9 mm, at which GPS/BDS-3 presents an improvement of approximately 31% for the

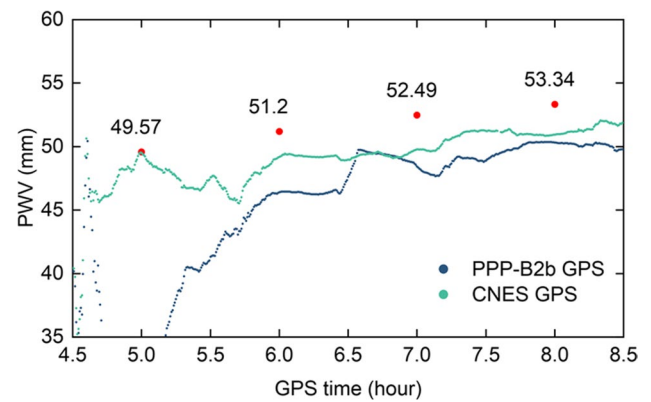


Fig. 16 PWV retrieval of GPS in a dynamic environment

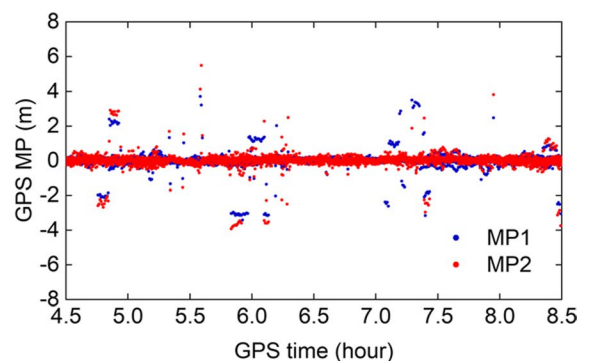
static experiment. The ZTD accuracy of PPP-B2b GPS is 26.4 mm, underperforming ERA and CNES-GPS by approximately 38% and 59%, respectively. For the ship-board experiments, the PWV of the PPP-B2b GPS/BDS-3 is consistent with the CNES results, and the difference between them is mostly within 1.5 mm after convergence.

Furthermore, our analysis shows that the accuracy of the real-time ZTD retrieval and the retrieval accuracy of CNES is more stable than that of ERA5 and PPP-B2b ("Comparison of the ZTD of PPP-B2b/CNES retrieved in static mode with ERA5" section). While the accuracy of ERA5 is affected by latitude, season, and other factors, the satellite clock accuracy of PPP-B2b GPS is lower than that of CNES. These led to the ZTD retrieval accuracy being lower than that of CNES. Furthermore, we found that the real-time T_m retrieved from the GPT3 model has a slight deviation from the T_m based on ERA5 ("Comparison of the PWV retrieved from a static mode of the PPP-B2b/CNES with ERA5" section). Using the PWV from ERA5 as a reference, the GPS/BDS-3 has a higher PWV retrieval accuracy than GPS only. Consequently, the PPP-B2b

Fig. 15 Shipboard receiver and GPS multipath in Houcha Bay



(a) Shipboard receiver

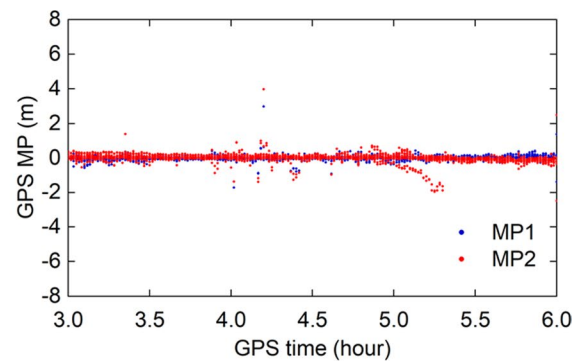


(b) GPS multipath

Fig. 17 Shipboard receiver and GPS multipath in Tangdao Bay



(a) Shipboard receiver



(b) GPS multipath

service provides a new way of real-time water vapor monitoring for remote areas without network links by broadcasting RTS via satellite.

Regarding the real-time PWV retrieval in the marine environments ("Retrieval experiments of dynamic shipboard data" section), we found that the retrieval performance and convergence time of GPS/BDS-3 of PPP-B2b were similar to those of CNES. Although this is a small study, the results show that the PPP-B2b service will have a promising application in marine meteorological research. At present, the retrieval accuracy of PPP-B2b is still slightly lower than that of CNES. Nevertheless, PPP-B2b is suitable for application in some environments where CNES is not performing well. As mentioned in "Introduction" section, no one has evaluated the PPP-B2b-based PWV over marine environments. Hence, the importance of our results using such satellite links lies both in their generality for navigation and their relatively good accuracy for retrieving environmental parameters.

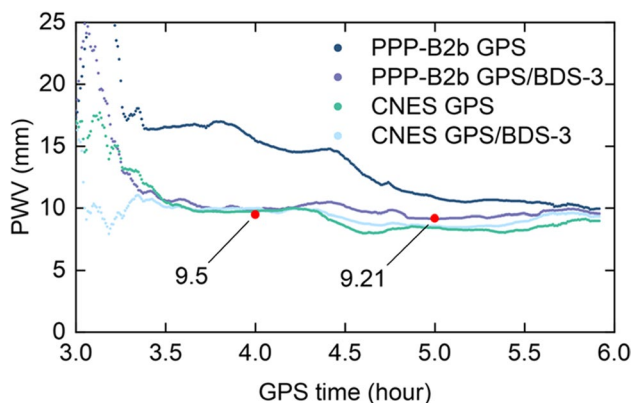


Fig. 18 PWV retrieval of GPS and GPS/BDS-3 in a dynamic environment

Conclusion

Since the BDS provides PPP-B2b service with strong anti-interference capability, users can receive RTS via GEO satellites, which expands the application range of real-time PPP for retrieving meteorological parameters, such as open sea and remote areas lacking network links. Therefore, this study evaluated the performance of PPP-B2b and CNES RTS in PWV retrieval and compared it with ERA5.

Experiments showed that the ZTD accuracy of ERA5 is 16.5 mm. The overall ZTD accuracy of PPP-B2b is slightly lower than that of CNES and ERA5, while the accuracy of GPS is 26.4 mm, and that of the GPS/BDS-3 combination is 17.4 mm. For PWV retrieval, the RMS of differences (RMSE) between PPP-B2b GPS and ERA5 is 4.9 mm, and that of the GPS/BDS-3 combination improves about 1.5 mm over GPS only. The PWV of CNES GPS/BDS-3 also can reach 2.3 mm. In dynamic experiments, the PWV of PPP-B2b GPS/BDS-3 can achieve similar results as CNES, and the difference is within 1.5 mm. There is a lack of sufficient data to test the long-term performance of the PPP-B2b service in dynamic PWV retrieval, which is the next step that needs to be taken.

According to the above results, the accuracy of PWV retrieval from the PPP-B2b service can meet the demand of monitoring requirements. As it will gradually support Galileo and GLONASS in the future, the accuracy of real-time retrieval will be further improved. It is expected that PPP-B2b will play an essential role in real-time weather research.

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Data availability The RINEX data of IGS stations are freely available at the following link <https://cddis.nasa.gov/archive/>. The real-time satellite ephemeris can be retrieved from: http://www.ppp-wizard.net/products/REAL_TIME/. The shipborne GNSS observations are available upon request to Mr. Hao Yang at upcyanghao@163.com.

Declarations

Competing interests The authors declare no competing interests.

Conflict of Interests The authors declare that they have no competing financial interests or personal relationships that could appear to influence the work reported in this paper.

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